BY ORDER OF THE SECRETARIES OF THE AIR FORCE, THE ARMY, THE NAVY, COMMANDANT OF THE MARINE CORPS, AND THE DIRECTOR OF THE DEFENSE LOGISTICS AGENCY

DEPARTMENT OF THE AIR FORCE MANUAL 24-210 AR700-143, NAVSUPINST4030.55E, MCO4030.40D DLAR4145.41

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Transportation

PACKAGING OF HAZARDOUS MATERIALS

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In accordance with the authority of DoDM 4140.01, Volume 9, DAFMAN 24-210/AR700-143/NAVSUPINST 4030.55E/MCO 37 4030.40D/DLAR 4145.41 establishes uniform guidance for packaging Hazardous Materials (HAZMAT) for safe, efficient, and legal storage, handling, and transportation, to include Department of Transportation Special Permit (DOT-SP), Competent Authority Approval (CAA), Certificate of Equivalency (COE) and Packaging Waivers for Military Air in accordance with Defense Transportation Regulation (DTR) 4500.9-R-Part II, Cargo Movement. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See DAFI 90-160, Publications and Forms Management, for a description of the authorities associated with the Tier numbers. Processes and authorities to waive requirements identified within this publication take precedence over Tier waiver authorities. This applies to Air Force units only. Ensure all Department of the Air Force (DAF) records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, Records Management and Information Governance Program, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the Office of Primary Responsibility

(OPR), using the DAF Form 847, Recommendation for Change of Publication; route DAF Form 847 from the field through the appropriate functional's chain of command.

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. It has been revised from DAFMAN 24-210/AR700-143/NAVSUPINST 4030.55F/MCO 37 4030.40D/DLAR 4145.41, *Packaging of Hazardous Material*, dated 17 March 2023. Changes include extensive updates to personnel and organizational data in **Attachment 7**. For doctrinal consistency, key terminology has been updated, with Department of Defense (DoD) being replaced with Department of War (DoW) and Surface Deployment and Distribution Command (SDDC) by Army Transportation Command (ARTRANS). Additionally, the update provides greater procedural and definitional clarity on the submission process for Certificates of Equivalency (COE).

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Chapter 1

GENERAL GUIDANCE

1.1. Applicability. This regulation is applicable to the Military Services and the Defense Logistics Agency (DLA), referred to as Department of War (DoW) components in this regulation. The term "Military Services," as used herein, refers to the Army, the Navy, the Air Force, the Marine Corps, and the Space Force.

1.1.1. In accordance with Title 49 Code of Federal Regulations (CFR), §173.7(a), DoW has the authority to certify for transport packagings equal to or greater in strength and efficiency than packagings meeting Title 49 CFR requirements. DoW also has the authority to reship any shippers certified shipment to any consignee provided the original packaging has not been damaged or altered in any manner.

1.1.2. Hazardous Materials (HAZMAT) packaging, which has been tested and passed the United Nations (UN) Performance Oriented Packaging (POP) standards, will be applied to HAZMAT for domestic shipments consistent with Title 49 CFR, Parts 100-185. **(T-0)**

1.1.3. DoW-managed HAZMAT shall be provided packaging protection at the lowest overall cost without compromising established DoW safety standards. Packaging shall provide adequate continuous protection to the packaged HAZMAT and shall prevent any release of the HAZMAT.

1.1.4. When HAZMAT is prepared for international shipment, the packaging and marking shall conform to the applicable modal regulations, i.e., International Civil Aviation Organization (ICAO) or International Maritime Organization (IMO)/International Maritime Dangerous Goods (IMDG) Code. The International Air Transport Association (IATA) Dangerous Goods Regulations (DGR) contains all ICAO Technical Instructions and is used by the airline industry; therefore, shippers must use the IATA DGR when shipping commercial air to ensure unique airline restrictions are adhered to. When shipping via military air, AFMAN 24-604/TM 38-250/NAVSUP PUB 505/MCO P4030.19/DLAI 4145.3, *Preparing Hazardous Materials for Military Air Shipments,* shall be followed. **(T-0)**

1.2. Internal Controls. Each DoW component will review their component specific packaging systems and procedures for compliance during hazardous material packaging field assistance visits, either with a technical assistance and operational review program, or separately, on an as-needed basis to evaluate the adequacy of field packaging operations, and conformance to this regulation. **(T-0)**

1.3. Transportation of Hazardous Materials. For transportation of hazardous materials on or off base, see DTR 4500.9R, Part II, Chapter 204, Section C.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. Commanders, Directors, or Heads of DoW Components.

2.1.1. Commanders, Directors, or Heads of DoW Components will comply with the policies, procedures, objectives, and guidelines in this regulation publication and DoD Manual 4140.01, DoD Manual 4140.70, DoD Manual 4140.65, AR 700- 15/NAVSUPINST 4030.28/AFMAN 24-206/MCO 4030.33/DLAR 4145.7, MIL-STD-129, MIL-STD-331, MIL-STD-648, MIL-STD-810, MIL-STD-1522, MIL-STD-1660, MIL-STD-1791, MIL-STD-2105 and ASTM D3951.

2.1.2. Designate individual Service focal points to coordinate the following:

2.1.2.1. Coordinate purchase requests for UN testing that are performed by a DOT- 124 approved third-party test facility.

2.1.2.2. Ensure that copies of all hazardous materials packaging test reports are forwarded to DLA Distribution as outlined in **paragraph 2.4.1**.

2.1.2.3. Develop internal operating procedures to handle organizational needs concerning proper HAZMAT packaging.

2.2. DoW Service Components Focal Points . DoW Service Components Focal Points will: participate in the DoW Hazardous Materials Packaging Working Group (HMPWG) as defined in **Attachment 5**. Service Focal Points jointly establish procedures and prepare any documentation necessary to implement this manual. Users contact their Service Focal Points for all clarifications and waivers. Service Focal Points are located in **Attachment 7**.

2.3. Assistant Secretary of War for Sustainment. The Assistant Secretary of War for Sustainment will, in accordance with Volume 9 of DoD Manual 4140.01, DoD Supply Chain Materiel Management Procedures, Material Programs, develop packaging programs for DoW materiel and oversees the Defense Packaging Policy Group. **(T-0)**

2.4. Defense Logistics Agency (DLA), (J344)

2.4.1. The Defense Logistics Agency will manage and maintain the POP Program, including pertinent information related to HAZMAT packaging testing. Activities will ensure that copies of all test reports, whether developed in-house or on behalf of the DoW by a DOT-approved third-party test facility, are forwarded to DLA Distribution, located at: DLA Distribution HQ J4 (POP Team), 430 Mifflin Ave, New Cumberland, PA 17070. (**T-0**)

2.4.2. Provide DLA Distribution J1 packaging training on the use of the POP Program.

2.5. Army Transportation Command (ARTRANS). ARTRANS will coordinate with the U.S. Department of Transportation to incorporate by reference this manual in Title 49 CFR §173.7. **(T-0)**

2.6. ARTRANS Safety Office. The ARTRANS Safety Office will maintain the SafetyNet web site available through the ARTRANS Electronic Transportation Portal. (TEAMS). **(T-0)**

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DoDM 4140.01, Volume 9, *DoD Supply Chain Materiel Management Procedures, Material Programs,* Latest Revision.

MIL-STD-129, *Standard Practice, Military Marking for Shipment and Storage,* Latest Revision

AFI 33-322, *Records Management and Information Governance Program,* Latest Revision.

AR 700-15/NAVSUPINST 4030.28D/AFMAN 24-206 IP/MCO 4030.33E/DLAR 4145.7. *Packaging of Materiel,* Latest Revision.

AFMAN 24-604/TM 38-250/NAVSUP PUB 505/MCO 4030.19K/DLAI 4145.3, *Preparing Hazardous Materials for Military Air Shipments,* Latest Revision.

TB 700-2, NAVSEAINST 8020.8C, TO 11A-1-47, *DoD Ammunition and Explosive Hazard Classification Procedures,* Latest Revision.

DLM 4000.25, Volume 2, *Supply Standards and Procedures, Chapter 17, Supply Discrepancy reporting,* Latest Revision.

Title 49 CFR Parts 100-185, *Transportation*

ASTM D4919-23, *Standard Specification for Testing of Hazardous Material Packagings* 168 *(Dangerous Goods) Packagings* Latest Revision.

Army Transportation Command (ARTRANS) *SafetyNet Program*

DI-L-3311 Explosive hazard classification data, Latest Revision

DI-PACK-80880 Transportability Report, Latest Revision

DI-L-1903 Part, Component or subsystem test plan, Latest Revision

DI-PACK-81059 Hazardous material POP test report format, Latest Revision

Prescribed Form

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

AFMAN—Air Force Manual

AMCOM—Aviation and Missile Command

APPX—Appendix

ARDEC—Armament Research, Development and Engineering Center

ARTRANS—Army Transportation Command

ASTM—American Society for Testing and Materials

CAA—Competent Authority Approval

CAGE—Commercial and Government Entity

CCN—Certification Control Number

CECOM—Communications-Electronics Command

CFR—Code of Federal Regulations

CIDS—Commercial Item Description

COE—Certificate of Equivalency

CONOPS—Concept of Operations

DGR—Dangerous Goods Regulation

DIDs—Data Item Descriptions

DLA—Defense Logistics Agency

DoD—Department of Defense

DoW—Department of War

DOT—Department of Transportation

DPPG—Defense Packaging Policy Group

DTR—Defense Transportation Regulation

DTS—Defense Transportation System

ECBC—Edgewood Chemical-Biological Center

EX Letter—Explosives Letter

EXECSEC—Executive Secretary

FHC—Final Hazard Classification

FMS—Foreign Military Sales

GFE—Government Furnished Equipment

HAZMAT—Hazardous Material

HMPWG—Hazardous Materials Packaging Working Group

HQ—Headquarters

IATA—International Air Transport Association

ICAO—International Civil Aviation Organization

IAW—In Accordance With

IBC—Intermediate Bulk Container

IHC—Interim Hazard Classification

IMDG—International Maritime Dangerous Goods

IMO—International Maritime Organization

JHCS—Joint Hazard Classification System

LP—Large Packaging

LTLP—Larger than Large Packaging

MOU—Memorandum of Understanding

OCONUS—Outside Continental United States

OPR—Office of Primary Responsibility

POP—Performance Oriented Packaging

PG—Packing Group

PPI—Provisional Packaging Instructions

POI—Program of Instruction

RH—Relative Humidity

SME—Subject Matter Expert

SP—Special Permit

SPI—Special Packaging Instruction

UN—United Nations

USADACS—US Army Defense Ammunition Center and School

Terms

Accountable Forms—Forms that the Air Force stringently controls, and which cannot be released to unauthorized personnel, since their misuse could jeopardize DoW security or result in fraudulent financial gain or claims against the government.

Certification—The act of confirming and verifying the completed package meets the requirements of the applicable modal regulation.

Certifier Preparer—One who physically recognizes the correctness of a package construction or has access to test data for that package and who then verifies in writing that it will perform to the level required. A certifier may perform one or more of the following acts of certification:

a. Performs a packaging operation in compliance with instructions prepared by a package designer.
b. Determines that the packaging and/or container has been manufactured, assembled, and marked in accordance with requirements.

Closure—A device which closes an opening in a receptable or a package.

Container—A single packaging, usually metal, intended as a long-life shipping and storage container for an item.

Excepted Quantities—Small amounts of a certain HAZMAT transported domestically by highway or rail that is not subject to all of the regulatory requirements of Title 49 CFR Part 173, when they meet the criteria of Title 49 CFR § 173.4(a).

Final Hazard Classification (FHC)—A Final Hazard Classification is either an EX-Letter or a JHCS Hazard Classification Printout, which defines the final hazard classification of an asset.

In-House Testing—HAZMAT packaging testing conducted at a DoW packaging test facility.

Larger than Large Packaging (LTLP)—A packaging that has no Title 49 CFR Packaging Standard, due to its size, which exceeds the maximum volumetric limit of 3 cubic meters for large packagings but otherwise meets the definition of a Large Packaging in Title 49 CFR § 171.8. (per Title 49 CFR § 173.7 Government operations and materials).

Level 3 Technical Data Package (or Product Design Data ref. MIL-STD-31000)—Engineering design data which describes the complete physical and/or performance characteristics of an item or component in sufficient detail to ensure that an item or component produced in accordance with this data will be essentially identical to the original item or component. For example, a final production-level packaging drawing (to include SPs), released for contractual use, for a long-life container.

Special Packaging Instruction—A SPI drawing illustrates packaging requirements for one item or a limited group of items of the same size and shape. SPI drawings detail special requirements for cushioning, blocking, bracing, and constructing containers. A bill of materials may also be included in an SPI drawing.

Attachment 2

PROCEDURES

A2.1. Contract Requirements. DoW components will specify packaging and labeling requirements in solicitations and contracts based upon UN POP unless the HAZMAT is excepted from the requirements in this Attachment or as directed by the DoW component Headquarters. It's vital that all worldwide and modal regulations are identified during procurement of HAZMAT. Explosive items procured must be properly classified under the UN system in order to ensure safe storage and shipment to CONUS and OCONUS points of use. These shipments must meet DoW/DOT hazard classification and packaging requirements as specified in referenced documents. **(T-0)**

A2.2. Multi-application HAZMAT Packaging. Containers designed for multiple inner components will be tested and certified to their maximum capacity as stated in the applicable regulation. This testing will be sufficient for the containers when loaded to less than their maximum capacity. Any void space must be completely filled with approved cushioning material to prevent their breakage or leakage and control their shifting within the outer packaging under conditions normally incident to transportation. Explosives containers with Level 3 Technical Data Packages, designed for multiple configurations of explosive articles, may be tested and certified to the configuration of maximum gross mass, irrespective of variations in inner article shape or orientation so long as these variables do not affect containment. **(T-0)**

A2.3. Marking of HAZMAT Packaging . Marking of HAZMAT packaging will be in accordance with the applicable modal regulation(s) and the latest revision of MIL-STD-129, *Military Marking for Shipment and Storage.*

A2.3.1. Packaging that successfully passes DoW hazardous materials packaging testing will be marked with the logo "USA/DoD". Activity-specific logos in **Attachment 6** may be used by DoW test activities if directed by the owner of the code. "USA/DoD" and activity specific logos may be added to large packaging performance markings.

A2.3.2. DoW components are authorized to mark the applicable DoD UN POP logo on Federal, Military or well established industrial (e.g., ASTM International) specification containers that pass UN POP testing (see **Attachment 6**). DoW may test vendor-manufactured packages and apply the DoD UN POP logo. Contractors, who perform the packaging and certification process for the DoW, may be authorized to use the DoD UN POP logo when stated in their contract.

A2.3.3. DoD logos and test reports may be provided to vendors, at the discretion of the procuring activity, only if the containers are Government Furnished Equipment (GFE), or if the containers are strictly controlled by configuration control drawings (including SPIs), first article tests are performed, and inspection procedures are followed by Government quality assurance personnel that validate compliance.

A2.3.4. In instances where the manufacturing date of a packaging cannot be determined on packaging design which has successfully passed performance testing, the year the packaging was filled may be used in lieu of the manufacturing date.

A2.3.5. English Language Marking Requirements. Manufacturer's packages, certified to UN specifications by a nation other than the United States, may be accepted into the DTS when the package markings are in English.

A2.4. HAZMAT Packaging Testing. Testing of HAZMAT packaging will be in accordance with Title 49 CFR Part 178 instructions authorized by the Competent Authority, and any differences designated by ICAO, IMDG, and IATA. **(T-0)** Use of ASTM D4919, *Standard Specification for Testing of Hazardous Materials (Dangerous Goods) Packagings,* is encouraged to ensure test repeatability. This publication can be found at **https://www.astm.org/**. HAZMAT packaging testing will be conducted according to the mode of transport, physical state of material, packing groups, weight, and container configuration rather than National Stock Numbers (NSN). **(T-0)** Any other HAZMAT that is within the test parameters may be shipped in that certified container.

A2.4.1. DoW components shall ensure that HAZMAT are in containers that have successfully passed the required test standards, unless that packaging is exempt from the requirement. **(T-0)** This may be accomplished by procuring the HAZMAT in certified containers, by performing in-house testing through the use of a DoW test facility, or by the use of a DOT- approved third-party contractor certified by DOT to perform UN Certified 3rd party testing of non-bulk POP. DoW components may require vendors to submit test reports and configuration control drawings when procuring HAZMAT in packaging conforming to UN specifications. Vendors may be asked to submit test reports by Data Item Description DI-PACK-81059 (Performance Oriented Packaging Test) or other applicable contractual packaging requirements. The decision to procure test data from a vendor is at the discretion of the Program Manager/Requirement Owner. An example of when procurement of vendor test data may be appropriate is when the packaging data for the item is also being procured. **(T-0)**

A2.4.2. HAZMAT packaging testing shall include all Title 49 CFR Part 178 requirements or the applicable Competent Authority decisions for the affected item's hazard class. **(T-0)** The following guidance applies to DoW component hazardous materials packaging testing:

A2.4.2.1. Prior to initiating a request for HAZMAT packaging testing, DoW components will research the DoD POP program to determine if a tested package/configuration already exists. **(T-0)**

A2.4.2.2. DoW components shall identify like items, e.g., acids, bases, etc. of HAZMAT and use a single test report to package and certify a related family to the maximum extent possible in accordance with the instructions published and authorized by the Competent Authority. **(T-0)**

A2.4.2.3. In order to qualify for air shipment, inner receptacles of a combination packaging containing liquids must be capable of withstanding without leakage an internal pressure differential standard and all other requirements of all air modal regulations, as applicable. Inner receptacles that do not meet this requirement must be packed in supplementary packaging that meets the pressure differential requirement, as well as all other requirements of the applicable modal regulation. **(T-0)**

A2.4.2.4. UN POP tests will be conducted on affected HAZMAT packaging, in addition to tests that may be required by the Federal or Military container specification. If the specification test requirements meet or exceed those required for UN testing specifications, the tests need not be duplicated if documented test results are available. UN POP standards tests will be conducted to the requirements of the most stringent mode of transportation anticipated. **(T-0)**

A2.4.2.5. Limited quantities, to be offered for air shipment, while not requiring UN Specification packaging must successfully pass tests required by ICAO Technical Instructions, Part 3, Chapter 4 and AFMAN 24-604, *Preparing Hazardous Materials for Military Air Shipments*, Attachment 19**. (T-0)**

A2.4.2.6. In lieu of using multiple samples of the same configuration in testing, the same packaging(s) (container(s) may be used to perform all required stack, drop, vibration, pressure, and any other required tests, provided that no damage or other changes have occurred to the packaging that would make it more likely to pass a test. If the same test package (packaging and contents, prepared as for shipment) passes one, but fails the next test, it is permissible to replace it with a test package of identical configuration, repeat the failed test, and continue testing.

A2.4.2.6.1. If a test package passes the first drop, but fails after any one of the following drops, it is permissible to replace it with an identical test package, repeat the failed drop, and continue testing. If, however, a package fails after only one drop, this constitutes a test failure.

A2.4.2.6.2. One sample packaging may be used to perform stack testing, provided that the test duration is a minimum of 72 hours. Packagings used in stack testing shall not be previously tested packagings. All stack tests shall be performed using empty packaging(s). Packagings that were used in stack testing (when empty) may be used as test samples in drop and vibration testing provided that no damage or other changes have occurred to the packaging that would make it more likely to pass a test. **(T-0)**

A2.4.2.6.3. One sample package may be used to perform vibration testing, provided that the test duration is a minimum of 3 hours. If a sample package, not previously used in testing, fails the vibration test, this constitutes a test failure.

A2.4.2.6.4. Packagings that have passed First Article testing may also be used to satisfy the multiple container test sample requirements by using a single packaging throughout all tests, under the above conditions.

A2.4.2.6.5. Periodic retesting of explosives containers with a Level 3 Technical Data Package. The package will undergo an initial periodic retest within the prescribed period in 49 CFR, with subsequent retesting at a maximum 5-year periodicity. The periodicity of the retest will be at the discretion of the responsible Military Service/Agency packaging authority. **(T-0)**

A2.4.2.6.6. If the capability exists to test a Larger than Large Packaging (LTLP) to the requirements of 49 CFR Part 178, Subpart Q, these tests may be performed as for an LP, and a large packaging UN marking may be applied to the packaging. The testing activity may use a vibration test method that is equally effective as 49 CFR Part 178.985, if necessary. In accordance with 49 CFR Part 178.910(a)(1)(ii), the LTLP shall be marked with the letter "W" immediately following the Large Packaging code letter. (T-0)

A2.4.2.6.7. DoD certification for one manufacturer does not apply to other manufacturers. Each manufacturer must obtain separate certification for their containers. (T-0)

A2.4.2.6.8. All POP reports submitted to the DoW Packaging Engineering Activity shall be in accordance with DI-PACK-81059.

A2.5. DoW Testing/Retesting. The following applies to hazardous materials packaging testing/retesting:

A2.5.1. The United States DOT is the US Competent Authority and considers all of DoW as one entity; (i.e., container manufacturer). Therefore, once a HAZMAT packaging configuration has been tested by one DoW activity, other DoW shippers do not need to test the same package configuration.

A2.5.2. For reparable hazardous items, the cognizant packaging design activity is responsible for the development of Special Packaging Instructions (SPI) or configuration control drawings and the required hazardous materials packaging testing of the reusable packaging.

A2.5.3. Periodic retesting will be accomplished on packaging configurations in production by DoW activities as required by 49 CFR § Part 178.601(e) or the Competent Authority. The Military Service/Agency having item management responsibility for the hazardous item is responsible for ensuring the periodic retesting is performed as required. (T-0)

A2.5.4. Specification Cylinder and UN Pressure Receptacle requalification and maintenance shall be accomplished IAW 49 CFR, Part 107, Subpart I, and Part 180, Subpart C. (T-0)

A2.6. DoW HAZMAT packaging test reports will not be provided to vendors except as otherwise provided in this manual. (T-0)

Attachment 3

CONTAINER USAGE

A3.1. Reuse/Refilling of Containers. The reuse of tested packages is authorized contingent upon the requirements of 49 CFR, Part 173.28. Reuse is authorized for domestic, international air, and military air. For international surface shipments, reusable fiberboard containers may not be used unless they are shipped inside intermodal transport containers.

A3.1.1. **Exempt Items.** The following materials are exempt from UN performance specification packaging test requirements. The packaging paragraph from AFMAN 24-604, Table A4.1. will specify required packaging. While UN specification packaging is not required, material may be subject to package performance tests.

Attachment 4

EXCEPTIONS AND SPECIAL APPROVALS

A4.1. Hazardous Materials Packaging Testing Exceptions. The following HAZMAT does not require testing to UN POP standards provided the provisions of the applicable modal regulations are met:

A4.1.1. Carbon Dioxide, Solid (Dry Ice).

A4.1.2. Magnetized material with a field strength of less than 0.00525 gauss at 4.527 meters (15 feet).

A4.1.3. Life Support Equipment.

A4.1.4. Class 2 Compressed Gas Cylinders.

A4.1.5. Class 7 Radioactive Material.

A4.1.6. Except for IBCs and LPs, a packaging with a capacity over 450L (119 gallons) as a receptacle for liquids or items weighing over 400 kg (882 pounds) and a capacity greater than 450L as a receptacle for solids.

A4.1.7. Limited quantities.

A4.1.8. Excepted quantities.

A4.1.9. Consumer commodities.

A4.1.10. Items, by modal regulations, require only a strong outer packaging.

A4.1.11. Larger than Large Packagings (LTLP). Containers, which exceed the 3 cubic meters limit for Large Packagings, and do not lend themselves to being practically subjected to performance testing due to excessive size, weight, or unique design characteristics, shall be qualified for shipment by the responsible Service's packaging authority. Qualification shall include testing to 49 CFR § 178.900 when possible, military container standards (e.g., MIL-STD- 1660 or MIL-STD-648), and engineering analysis that demonstrates the adequacy of the shipping configuration to withstand forces normally incident to the intended mode(s) of transportation and shall satisfy the applicable requirements of Title 49 CFR §173.24 and §178.900. **(T-0)**

A4.1.12. Large & robust articles. If the responsible Service Safety Office classifies an item as a large & robust explosive on an IHC or FHC, the item may be transported unpackaged IAW the conditions of Title 49 CFR §173.60(b)(14). A COE will be issued by the responsible Service for large & robust items with an IHC. A container, cradle, or other suitable handling device must demonstrate by testing and or analysis, that the item will not become loose during normal conditions of transport and is in accordance with DoW-approved procedures. Approved containers for items with a FHC will be identified by a packaging note. **(T-0)**

A4.2. Class 1 (explosives) Exceptions. Class 1 (explosives) Exceptions. Class 1 (explosives) materials owned by the DoW, packaged prior to January 1, 1990, as described in Title 49 CFR § 173.7(e), are excepted from the UN POP standards of 49 CFR §173.62 and the associated packaging and marking requirements of Title 49 CFR Part 178. The packaging must have maintained its integrity. Shipping papers must identify the material as government owned goods packaged prior to 1990 as required by the Defense Transportation Regulation (DTR), 4500.9R, Part II. Marking and labeling requirements of the most current Title 49 CFR Part 172 must be complied with except that marking boards may be used for palletized loads in lieu of remarking and re-labeling each individual item/package. When an individual item/package is removed from the palletized load and offered for transport, it must be marked and labeled as required by Title 49 CFR Part 172. **(T-0)**

A4.3. Provisional Packaging Instructions (PPI). A PPI is the detailed packing instructions in accordance with Title 49 CFR specifically designed to package a small quantity of particular explosive/ammunition item that has not had a final hazard classification assigned during the research and development phase for domestic shipments. Only the designated DoW packaging research and development activities listed in **Attachment 6** are authorized to prepare and approve PPIs. A PPI may be required when applying for an interim hazard classification for an explosive/ammunition item in the research and development phase.

A4.4. DOT-Special Permits (DOT¬SP). A DOT-SP grants administrative authority to deviate from specific requirements, e.g. alternative packaging, testing procedures, etc. of Title 49 CFR. The government or commercial shipping activities must submit DOT-SP requests to the DoW Service/Agency focal point listed in **Attachment 7** who, in turn, will submit the request to the SDDC Safety Office, (AMSD-SA). The application requirements for submitting a DOT-SP are specified in Title 49 CFR § 107.105 and requirements outlined in **Attachment 8**. Shipping papers shall be annotated with the DOT-SP number. If required by the DOT-SP, the number shall also be marked on shipping container. DOT-SP number must be marked on the shipping container for Hazard Class 1. A copy of the DOT-SP shall accompany the shipment. DOT-SPs may be used for international shipments if the modal regulation is listed in regulations exempt from paragraph of the DOT-SP. A new DOT-SP application requires a minimum of 120 days for routine DOT processing and 30 days for an emergency DOT-SP. The holder of the DOT-SP must file for a renewal no less than 60 days prior to expiration of the existing DOT-SP in accordance with 49 CFR § 107.109. Approved DOT-SPs are available on the DOT web site or SDDC SafetyNet. In accordance with DTR 4500.9-R, Part II, Chapter 204, a quarterly usage report of DOT-SPs must be submitted to the Service/Agency Component focal point. **(T-0)**

A4.5. Competent Authority Approvals (CAA). A CAA is a written approval for specific hazardous material which by modal directive requires approval of the hazard classification or the packaging by a National Competent Authority prior to shipment. The shipping activity that requires a packaging CAA must forward the CAA request to their DoW Military Service/Agency focal point listed in **Attachment 7** containing the information outlined in **Attachment 9**. A CAA issued for international shipments may also be used for domestic shipments. SDDC shall maintain copies of all CAAs in an electronic file in the SafetyNet System. A CAA issued for approval of a hazard classification may also be used as a packaging approval if the packaging description/method is included in the document. Shipping papers shall be annotated with the CAA number. If required by the CAA, the number shall also be marked on the shipping container. CAA must be marked on shipping containers for Hazard Class 1. A CAA requires a minimum of 120

days for routine processing and 30 days for an emergency CAA (See **Attachment 9**). The holder of the CAA must file for a renewal no less than 60 days prior to expiration of the existing CAA in accordance with Title 49 CFR § 107.705. A copy of the CAA shall accompany all shipments. In accordance with DTR 4500.9-R, Part II, Chap 204, a quarterly usage report of CAAs must be submitted to the Service/Agency Component focal point. Explosive approval letters with packaging notes are considered CAA's. **(T-0)**

A4.6. Certificate of Equivalency (COE). A COE is a certification issued by DoW to authorize transport of HAZMAT (including ammunition and explosives) using packaging that differ from the prescribed regulations in Title 49 CFR. A COE certifies that the selected packaging design meets or exceeds the requirements of Title 49 CFR for the commodity being shipped. A COE shall specify the approved packaging within the document or refer to separate packaging document (PPI, SPI, drawings, etc.).Requests for COE's will be submitted to DoW Service/Agency focal points or packaging lab listed **in Attachment 7**. Once the focal points or packaging lab has all the necessary information, they will review the completed Hazardous Material Data Package. The COE's approval will be conducted by the services focal point or COE POC (or designate). Authorized COE writers are appointed in writing by the appropriate service authority and are listed in the DTR 4500.9R, Part II, Chapter 204. Once the COE is signed and released, there are no modifications to the original copy authorized. If there are any changes or additions to items listed in the COE, a new COE must be released after all required testing (if required) has been completed. The following information applies to COEs:

A4.6.1. COE is required, but not limited to, the below general situations:

A4.6.1.1. Packaging instruction per Title 49 CFR is specified, but the packaging used is different, or packaging used is different than specified by a packaging instruction.

A4.6.1.2. The United Nations Identification Number listed in the 49 CFR and assigned by an authorized DoW Safety Office on an IHC, or on the DOT Explosive Letter, corresponds to packaging instruction "PI-101" for explosives (see 49 CFR § 173.62, Tables 1 and 2.).

A4.6.1.3. Any item classified by the DoW on an IHC if the specified packaging differs from Title 49 CFR and cannot be issued a CAA or DOT-SP.

A4.6.1.4. The item contains compressed gas cylinder that is not constructed and/or tested per Title 49 CFR standards.

A4.6.1.5. Packagings that exceed the 3 cubic meters volume limit for Large Packagings, qualified for use per A4.1.11.

A4.6.1.6. Large and robust explosive articles as defined in **paragraph A4.1.12** of this regulation.

A4.6.1.7. A COE is not required when the packaging deviates from Title 49 CFR if the package has a DOT-SP, CAA or if the packaging has been approved and listed on the DOT EX letter in the notes. In this case, the DOT-SP, CAA or EX number issued by the DOT Competency Authority should be used. The exception would be when these documents specifically refer to the use of a COE in conjunction with them.

A4.7. COEs are used on a limited basis for domestic shipments of HAZMAT. A COE is authorized for use on international shipments if moved on a military controlled airlift, (i.e.), DoW chartered commercial aircraft under a DoW contract, and the movement is between DoW co-located activities to include operations in theater locations. A COE will not otherwise be acceptable for use to transport an item in a host country.

A4.8. A request for a COE must include the requirements defined in Attachment 10. Examples of supporting data required for a COE are also included in **Attachment 10.** A Certification Control Number (CCN) will be issued to identify the service issuing the COE using the following prefixes: **(T-0)**

A4.8.1. DLA−DH

A4.8.2. Army−AY

A4.8.3. Air Force−AF

A4.8.4. Navy−NA

A4.8.5. Marine Corps−MC

A4.8.6. Combine these prefixes with the current calendar year to construct the CCN; for example, DH 12-XXX. The CCN must be annotated on shipping papers according to the DTR, Part II and the exterior of the containers. A copy of the COE must also accompany the shipment. **(T-0)**

A4.9. Packaging Waiver for Military Air. DoW Services/Agency focal points listed in **Attachment 7** may issue packaging waivers for DoW military-controlled airlift in accordance with AFMAN 24-604/TM 38-250/NAVSUP PUB 505/MCO 4030.19K W/ERRATUM/DLAI 4145.3, *Preparing Hazardous Materials for Military Air Shipments*.

A4.10. Opening Non-Bulk Packages for Inspection. DoW components may open, inspect and subsequently reship packages only when the configuration of HAZMAT remains unchanged and the package is closed using manufacturer or shipper provided instructions. See AFMAN 24-604 for additional restrictions for military airlift.

A4.11. Repackaging. Items removed from original UN POP for distribution in different quantities or packaging configurations must be packaged using the DoD POP program in conjunction with a Special Packaging Instruction (SPI), the applicable National Stock Number (NSN) packaging drawing or approved packaging drawing issued by the responsible service. DoW shippers may apply UN Specification markings specified in the instruction being used to prepare or package the HAZMAT. When commercial packaging (authorized for the material being packaged) is used, the manufacturer provided packing or closing instructions must be used. Services and DoW agencies may not repackage an item IAW a CAA or DOT-SP unless the DoW is a party to the CAA or DOT- SP. Ammunition items shall be packaged IAW the applicable NSN item drawing in order to maintain compliance with the assigned DoW Final Hazard Classification. **(T-0)**

A4.12. Noncompliant Hazardous Materials Packaging.

A4.12.1. New Procurement. When DoW components determine that a vendor's certified HAZMAT packaging does not comply with contractual HAZMAT packaging requirements, suspend the receipt in material condition code "L." Follow the procedures in DLM 4000.25-M, Defense Logistics Management System (DLMS), Volume 2 (Supply Standards and Procedures) AP2.5-2 and promptly notify the applicable procuring activity. The procuring activity will coordinate a review of the discrepancy report and provide written disposition instructions to the activity holding the discrepant material. **(T-0)**

A4.12.2. Station Returns and Receipts from Other DoW Activities. When DoW components determine that another DoW Activity's certified HAZMAT packaging does not comply with Title 49 CFR and/or modal requirements, suspend the receipt in material "J" (suspended stock) or material condition code "K" (suspended returns) as outlined in DLM 4000.25, Volume 2, AP2.5-2. Promptly notify the applicable managing activity of suspended material. The managing activity will coordinate a review of the discrepancy report and provide written disposition instructions to the activity holding the discrepant material. **(T-0)**

A4.12.3. Foreign Military Sales (FMS). Approved COE/CAA documents shall not be utilized by foreign nation(s) to return items to CONUS for weapon system modifications or repairs. The foreign nation assumes ownership and responsibility for acquiring proper HAZMAT transport documents after accepting final delivery of weapon system(s). **Note:** A COE is not valid for movement of HAZMAT over international roads or via overseas commercial air. **(T-0)**

Attachment 5

DOW HAZARDOUS MATERIAL PACKAGING WORKING GROUP CHARTER

A5.1. Objective. The DoW Hazardous Material Packaging Working Group (HMPWG) is a sub-group of the Defense Packaging Policy Group (DPPG) authorized by the DPPG Charter. The HMPWG is a decision-making team responsible for developing, recommending, and implementing changes to ensure the most efficient procedures, policy, and standardization of hazardous material packaging throughout the Military Services and Defense Agencies.

A5.2. Functions. The HMPWG will develop and coordinate hazardous material packaging policy. The HMPWG members will provide and exchange information to modify/establish policies and procedures for the most effective use of DoW resources. Recommendations will be forwarded to the DPPG or applicable Service/DLA headquarters as required. Areas of primary interest include:

A5.2.1. Hazardous material packaging. Modify or establish hazardous material packaging policies and procedures to meet military operational requirements while ensuring compliance with applicable modal regulatory directives. Ensure individual Component policies align to require proper packaging and preparation methods for the safe and efficient storage, handling, and transport of hazardous materials.

A5.2.2. Testing. Establish testing procedures to ensure UN POP standards are met for Service/DLA managed items. Coordinate testing to prevent redundancy and provide the most efficient use of testing resources. Support and advise the DoD POP program as it pertains to each Service's requirements.

A5.2.3. Compliance. Review and recommend changes to regulatory requirements. Develop and coordinate recommendations to ensure compliance with DoW and commercial modal regulations.

A5.2.4. Training and Course Development. Evaluate and approve proposed training courses, curriculum, delivery, and resources on the preparation and shipment of hazardous materials. See **Attachment 11** for further details.

A5.2.5. Communication. Develop effective means to distribute hazardous material packaging requirements to operational personnel.

A5.2.6. Process Improvement. Strive for continuous improvement through implementation of new packaging materials, processes, and procedures.

A5.3. Procedures. The HMPWG will work together to ensure safe packaging of hazardous materials throughout the logistics cycle.

A5.3.1. Organization. The HMPWG will consist of primary and advisory members. Primary members will include packaging policy representatives from each Military Service and DLA. Members of the HMPWG will rotate duties as executive secretary (EXECSEC) and then chair every two years in the order listed below. After serving a 2-year period as the EXECSEC, the EXECSEC will assume the duties of Chair for a 2-year period. In the absence of the Chair, the EXECSEC will assume the Chair's duties during HMPWG meetings. Advisory members will include General Services Administration, DLA Distribution and representatives of activities performing hazardous materials packaging testing. **(T-0)**

A5.3.1.1. Primary Members

A5.3.1.1.1. Air Force Materiel Command (AFMC), Transportation and Packaging Policy.

A5.3.1.1.2. Army Sustainment Command Packaging, Storage, and Containerization Center (ASC-PSCC).

A5.3.1.1.3. United States Marine Corps, Deputy Commandant for Installation and Logistics.

A5.3.1.1.4. Naval Supply Systems Command, Weapon Systems Support.

A5.3.1.1.5. DLA Logistics Operations (J3).

A5.3.2. Guests. Members are responsible for the invitation of their respective Military Department and/or Agency guests. To maintain the effectiveness of the HMPWG, guests should be limited to those who may contribute significantly to the established agenda. Guest attendance is subject to approval by the Chair, or the EXECSEC in the absence of the Chair.

A5.3.3. Meeting. The HMPWG will as a minimum meet annually during the fourth quarter (FY) or as required by determination of the members. The Chair, in coordination with the primary members, will designate the dates and location of the meeting. **(T-0)**

A5.3.4. Agenda. Members will provide proposed agenda topics, with talking papers, to the Chair and EXECSEC by the designated time established by the meeting announcement. The EXECSEC will develop and distribute a final agenda with supporting talking papers to members prior to the meeting. **(T-0)**

A5.3.5. Travel Funds. Participating organizations provide travel funds for members to attend HMPWG meetings.

Attachment 6

PERFORMANCE ORIENTED PACKAGING IDENTIFICATION CODES

A6.1. Package Design Activity Codes. These codes are for the identification of the activities listed and are not to be applied to any packaging unless directed by the specific activity represented by the code.

Table A6.1. Packaging Design Activity Code.

Defense Logistics Agency Fort Belvoir, VA	USA/DoW/DLA
Defense Logistics Agency- Land & Maritime Columbus, OH (CAGE 16236 & 14933)	USA/DoW/DLC
Defense Logistics Agency- Energy Fort Belvoir, VA (CAGE 52838)	USA/DoW/DLF
Defense Logistics Agency- Aviation Richmond, VA (CAGE 13873)	USA/DoW/DLG
Defense Logistics Agency- Troop Support Philadelphia, PA (CAGE 14153)	USA/DoW/DLI
Armament, Research, Development and Engineering Center (ARDEC) (RDAR-EIL-P) Picatinny Arsenal, NJ (CAGE 19200)	USA/DoW/AYD
Armament, Research, Development and Engineering Center (ARDEC) (RDAR-EIL-TP) Rock Island, IL (CAGE 59678)	USA/DoW/AYA
Army Edgewood Chemical-Biological Center (ECBC) (AMSRD-ECB-END) Rock Island IL, (Cage 5B5M3)	USA/DoW/AYR
Army Edgewood Chemical-Biological Center (ECBC) (AMSRD-ECB-ENA-P) Aberdeen Proving Ground, MD (Cage 39KG1)	USA/DoW/AYE
Army Defense Ammunition Center and School (USADACS) McAlester AAP, OK (CAGE 28620)	USA/DoW/DEV
Army Communications-Electronics Command (CECOM) Aberdeen Proving Ground, MD (CAGE 80063)	USA/DoW/AYC
Army Medical Materiel Agency Frederick, MD (CAGE 66732)	USA/DoW/AYS
Soldier Systems Center Natick, MA	USA/DoW/AYN
Army Aviation and Missile Command (AMCOM) Huntsville, AL (CAGE 18876)	USA/DoW/AYM
Army Logistics Support Activity, Packaging, Storage and Containerization Center, Tobyhanna, PA	USA/DoW/AYP
Air Armament Complex Eglin AFB, FL (CAGE 32231)	USA/DoW/AF18
Rome Research Site Griffiss AFB, NY (CAGE 07877)	USA/DoW/AF17
Space and Missile Systems Center Los Angeles AFB, CA (CAGE 07868)	USA/DoW/AF19

Ogden Air Logistics Complex Hill AFB, UT (CAGE 98747)	USA/DoW/AF70
Oklahoma City Air Logistics Complex Tinker AFB, OK (CAGE 98748)	USA/DoW/AF71
Warner Robins Air Logistics Complex Robins AFB, GA (CAGE 98752)	USA/DoW/AF84
AF Packaging Technology and Engineering Facility Wright-Patterson, AFB, OH (CAGE 0B275)	USA/DoW/AF69
Marine Corps, Washington, DC (CAGE 80372)	USA/DoW/MCH
Marine Corps Research, Development and Acquisition Command Quantico, VA (CAGE 5N998)	USA/DoW/MCQ
Naval Air Systems Command Washington, DC (CAGE 30003)	USA/DoW/NAA
Naval Information Warfare Systems Command (NAVWAR) Washington, DC	USA/DoW/NAB
Naval Facilities Engineering Command Washington, DC (CAGE 80091)	USA/DoW/NAC
Naval Sea Systems Command Washington, DC (CAGE 53711)	USA/DoW/NAD
Naval Supply Systems Command/Weapon Systems Support (RIC NRP) Philadelphia, PA (CAGE 80132)	USA/DoW/NAE
Naval Supply Systems Command/Weapon Systems Support (RIC NRP) Mechanicsburg, PA (CAGE 03950)	USA/DoW/NAF

Attachment 7

RESPECTIVE MILITARY SERVICE/AGENCY WHEN SUBMITTING REQUESTS FOR CERTIFICATE OF EQUIVALENCIES (COE), DOT-SPECIAL PERMITS (DOT- SP), AND COMPETENT AUTHORITY APPROVALS (CAA)

A7.1. When a DOT-SP, CAA, COE or hazard classification is required. The managing activity for the affected item will prepare the request and submit it to the military service/agency focal point listed in **Table A7.1**. For a hazard classification contact the respective military service/agency identified below: **(T-0)**

A7.1.1. When a COE is required, see A4.6.

A7.2. COE and interim hazard classification (IHC) procedures. A COE requires documented (IHC) information for the packaged item(s). If a packaging DOT-SP or CAA is required, the requester must prepare the package identified in **Attachment 10**. The request must include a copy of the IHC, and supportive data used to determine that classification. The requester must validate the need for an IHC and indicate why the final hazard classification has not been established. **(T-0)** DoW agencies listed in the department of defense explosives classification procedures (TB 700-2, NAVSEAINST 8020.8C, TO 11A-1-47) are authorized to assign interim hazard classification (IHC) to allow the transportation of explosive materials that has not been assigned a final hazard classification. An IHC will be used for a COE until a final classification is established (in some cases, the DOT will not issue a packaging CAA or DOT-SP until a final hazard classification is assigned). See **Attachment 10** for the procedures to request a COE.

Table A7.1. Service/Agency Addresses.

AIR FORCE/SPACE FORCE	
COE/CAA/DOT-SP HQ AFMC/A4RT 4375 Chidlaw Rd Wright Patterson AFB, OH 45433 Attn: Mr. Ken Erhart or Mr. Jack Roberts Commercial: 937-257-4503 / 937-904-6524 DSN: 312-257-4503 Email: kenneth.erhart.1@us.af.mil Email: jack.roberts.5@us.af.mil **AIR FORCE PACKAGING TECHNOLOGY & ENGINEERING FACILITY** 5215 Thurlow Street, Bldg. 70 Wright-Patterson AFB, OH 45433 Attn: Mr. Joel Sullivan Commercial: 937-257-3362 DSN: 312-257-3362 Email: joel.sullivan@us.af.mil	**Explosive Classification CAAs/DOT-SP** Explosive Classification CAAs/DOT-SP HQ AFSEC/SEWC 9700 G Ave SE Kirtland AFB, NM 87117-2662 Attn: Mr. Paul Wever & Mr. Allen Kelly Commercial: 505-846-1386 /6058 DSN: 312-246-1386/6058 Email: paul.wever.5@us.af.mil Email: allen.kelly.4@us.af.mil

ARMY	
Hazmat Focal Point U.S. Army Sustainment Command Packaging, Storage, and Containerization Center Mr. Jacob Gogno or Ms. Nisha Shreshtha AMAS-SPI-P 11 HAP Arnold Blvd. Tobyhanna, PA 18466-5097 Commercial: (570) 615-8845 Email: jacob.b.gogno.civ@mail.mil Email: Nisha.shreshtha.2.civ@army.mil	Explosive Hazard Classification Defense Ammunition Center U.S. Army Technical Center for Explosive Safety ATTN: ATCL-ESE 1 C Tree Road, Building 35 McAlester, OK 74501-9053 Email: mcalester.usamc.list.dac-hc@army.mil
Missiles and Related COEs U.S. Army Aviation and Missile Command ATTN: AMAM-LMP-T Mr. Willie Sharp Building 5302, 2nd Floor Redstone Arsenal, AL 35898-5110 DSN: 746-0089 Commercial: (256) 876-0089 Email: willie.h.sharp2.civ@army.mil	COE/PPI for Ammunition/Explosives U.S. Army Combat Capabilities Development Command Armaments Center Mr. Steven Sicoli ATTN: FCDD-ACE-LP Picatinny Arsenal, NJ 07806-5000 Commercial: (520) 684-1984 Email: steven.r.sicoli.civ@army.mil U.S. Army Combat Capabilities Development Command Armaments Center Mr. Joseph Hanssen Rock Island Arsenal, IL 61299-5000 ATTN: FCDD-ACE-LTP Commercial: (520) 714-1653 Email: joseph.w.hanssen.civ@army.mil
NAVY	
COE/CAAs (Ordnance)/DOT-SP Director Naval Surface Warfare Center Indian Head Division Picatinny Detachment Code G1 Michael Kelly (Primary), Brendan Gibson (Alternate), and Stephen Perlas (Alternate) Picatinny Arsenal, NJ 07806- 5000 Commercial: 973-724-3388 DSN: 880-3388 Email: michael.l.kelly1.civ@us.navy.mil Email: brendan.k.gibson.civ@us.navy.mil Email: Stephen.l.perlas.civ@us.navy.mil	COE/CAA/DOT-SP NAVSUP Weapon Systems Support Code N242 – Susan Starks 5450 Carlisle Pike Mechanicsburg, PA 17055 Commercial: 717-605-5520 DSN: 430-5520 Email: susan.e.starks.civ@us.navy.mil Email: navsupwssmech.hazmat@us.navy.mil

Naval Ordnance Safety and Security Activity Attn: Code N85 B Farragut Hall 3817 Strauss Avenue, Suite 108 Indian Head, MD 20640-5151 Commercial: (240) 435-1272 Email: gayland.e.walseman.civ@us.navy.mil	
U.S. MARINE CORPS	
COE/CAA/DOT-SP Commandant of the Marine Corps I&L Code LPD-1 Attn: Mr. Lakye Franklin Headquarters, United States Marine Corps, Pentagon Rm 2E211 Washington, D.C. 20350-3000 Commercial: (571) 256-2763 Email: lakye.franklin@usmc.mil	Explosive to Hazard Classification CAA/DOT-SP Commanding Officer Naval Ordnance Safety and Security Activity Attn: Code N85B Farragut Hall 3817 Strauss Ave, Suite 108 Indian Head, MD 20640-5151 Commercial: (240) 435-1272 Email: gayland.e.walseman.civ@us.navy.mil
DEFENSE LOGISTICS AGENCY (DLA)	
COE/CAA/DOT-SP Defense Logistics Agency Attn: J344 (Mr. Eric Prowell) (Primary) 8725 John J Kingman Rd. Ft. Belvoir, VA 22070 Commercial: (717) 961-9726 Email: eric.prowell@dla.mil Alternate Email: dla_pkg_sr@dla.mil	COE/CAA/DOT-SP Defense Logistics Agency Attn: J344 (Ms. Jennifer Smith) (Alternate) 430 Mifflin Ave. Room 1249 New Cumberland, PA 17070 Commercial: (717) 770-4492 Email: jennifer.smith2@dla.mil
ARMY TRANSPORTATION COMMAND (ARTRANS)	
Attn: AMSSD-SA (Mr. Jeffrey M. Leitschuh or Mr. Thomas M. Kudasz) 1 Soldier Way Building 1900W Scott AFB, IL 62225-5006 Email: jeffrey.m.leitschuh.civ@army.mil Commercial (TEAMS): (520) 944-7031 Email: thomas.m.kudasz.civ@army.mil Commercial (TEAMS): (520) 944-6660	

Attachment 8

PROCEDURES TO SUBMIT DOT-SP

A8.1. General Procedures. Comply with Title 49 CFR §107.105 and the requirements below when requesting a DOT-SP. Requests for a packaging CAA or DOT-SP for Class 1 items must be directed to the Service focal points in **Table A7.1**. The focal points will work with HQ SDDC and the DoW Explosive Safety Board to obtain a CAA or DOT-SP from DOT. However, when contingency or mission critical operations exist, the HQ Military Services reserve the right to work directly with DOT. **(T-0)**

A8.1.1. Shippers who have authority to upload data into SafetyNet may submit a request for a DOT-SP or forward a request to their Service/Agency focal point (item manager). All documents defined in A8.2 must be uploaded in the required SafetyNet fields. Any request that does not contain all of the required documents will be rejected and returned to the requestor for resubmission.

A8.1.2. Each request will include a cover letter, which states why the DOT-SP is being requested and identifies the mode(s) of transportation affected. **(T-0)**

A8.2. Data Package. Collect the supporting technical data required from the appropriate sources (i.e., manufacturer, program offices, etc.) and attach these data as **Attachments 1** through **Attachment 7** described below as part of requesting a DOT-SP. The supporting technical data accompanies the request. NOTE: When the information requested does not apply, write N/A on that line, e.g., 'Actual Item Drawings – N/A.'

A8.3. Example of Completed Package.

A8.3.1. **Attachment 1** −Information to Support Special Permit:

A8.3.1.1. Product Nomenclature, National Stock Number(s), Part Number(s), and EX-number (if assigned).

A8.3.1.2. Hazard Class/Division. For Class 1 material, include the Storage Compatibility Group.

A8.3.1.3. UN Identification Number.

A8.3.1.4. UN Proper Shipping Name (ICAO and IMDG).

A8.3.1.5. Item Description and Drawing Number. Include the item net and gross weight. For Class 1 material, also include the net explosive weight.

A8.3.1.6. Packaging Description and Instructions/Drawing Numbers. If a COE and/or DOT-SP have been issued, include the number and description.

A8.3.1.7. Include COE transport documents, if applicable.

A8.3.1.8. Difference between DOT and DoW containers.

A8.3.1.9. Reports of Test(s) Conducted. List and provide a short description of the tests conducted on the container (i.e., MIL-STD-648, FED-STD-101C, MIL-R-8583A, Hazardous Materials Packaging Tests, Transportation and Handling Vibration Test, etc.). If POP tests or equivalent tests have been conducted, provide report with POP marking container.

A8.3.2. **Attachment 2** −Approved item drawings.

A8.3.3. **Attachment 3** −Approved item test reports.

A8.3.4. **Attachment 4** −Approved container drawings.

A8.3.5. **Attachment 5** −Approved container test.

A8.3.6. **Attachment 6** −Certification of Equivalency (if applicable).

A8.3.7. **Attachment 7** −Hazard Classification (if applicable).

A8.3.8. Submit a usage report quarterly to SDDC including all DOT-SP renewal requests.

A8.3.9. **Figure A8.1** depicts an approved DOT-SP from the Department of Transportation.

Figure A8.1. Special Permit Sample - Remainder of documents can be viewed on SafetyNet.



U.S. Department
of Transportation

Pipeline and Hazardous Materials Safety Administration

1200 New Jersey Avenue, SE
Washington, DC 20590

SPECIAL PERMIT AUTHORIZATION

DOT-SP 20556

EXPIRATION DATE: 2027-06-30

GRANTEE: Department of Defense, Military Surface Deployment & Distribution Command
Scott AFB, IL

In response to your June 27, 2025, application for party status to DOT-SP 20556, Department of Defense, Military Surface Deployment & Distribution Command is hereby granted party status to DOT-SP 20556 as a shipper only in accordance with 49 CFR 107.113.

Copies of this special permit may be obtained by accessing the Office of Hazardous Materials Safety Homepage at https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search. The most recent revision of the special permit supersedes all previous revisions of the special permit. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

If you have questions regarding this action please call the Office of Hazardous Materials Safety, General Approvals and Permits Branch at (202) 366-4535.

Issued in Washington D.C. on **July 11, 2025.**

for William Quade
Acting Associate Administrator for Hazardous Materials Safety

Tracking Number: 2025074054

DUNS Number on file: 080271555

Attachment 9

PROCEDURES TO SUBMIT COMPETENT AUTHORITY APPROVAL (CAA)

A9.1. General Procedures. Comply with 49 CFR Part 107 Subpart H and the requirements below when requesting a CAA. The managing activity for the affected item will prepare the request and submit all documents in A9.2 to the applicable Military Service/Agency focal point listed in **Attachment 7**. Military Service/Agency focal points will submit completed packages to SDDC for submission to DOT. (See **Figure A9.1** for Sample). **(T-0)**

A9.2. Data Package. Collect the supporting technical data required from the appropriate sources (i.e., manufacturer, program offices, etc.) and attach these data as **Attachments 1** through **Attachment 8** described below as part of requesting a CAA. The supporting technical data accompanies the request.

A9.3. Example of Completed Package. When the information requested does not apply, write N/A on that line, e.g., 'Actual Item Drawings – N/A.'

A9.3.1. **Attachment 1** - Information to Support Competent Authority Approval:

A9.3.1.1. Product Nomenclature, National Stock Number(s), Part Number(s), and EX-number (if assigned).

A9.3.1.2. Hazard Class/Division. For Class 1 material, include the Storage Compatibility Group.

A9.3.1.3. UN Identification Number.

A9.3.1.4. UN Proper Shipping Name (ICAO and IMDG).

A9.3.1.5. Item Description and Drawing Number. Include the item net and gross weight. For Class 1 material, also include the net explosive weight.

A9.3.1.6. Packaging Description and Instructions/Drawing Numbers. If a COE and/or a DOT-SP have been issued, include the number and description.

A9.3.1.7. Difference between DOT and DoW containers.

A9.3.1.8. Reports of Test(s) Conducted. List and provide a short description of the tests conducted on the container (i.e., MIL-STD-648, FED-STD-101C, MIL-R-8583A, Hazardous Materials Packaging Tests, Transportation and Handling Vibration Test, etc.).

A9.3.1.9. **Attachment 2** −Approved item drawings.

A9.3.1.10. **Attachment 3** −Approved item test reports.

A9.3.1.11. **Attachment 4** −Approved container drawings.

A9.3.1.12. **Attachment 5** −Approved container test reports.

A9.3.1.13. **Attachment 6** −Department of Transportation Special Permit (if applicable).

A9.3.1.14. **Attachment 7** −Certification of Equivalency (if applicable).

A9.3.1.15. **Attachment 8** −Hazard Classification (if applicable).

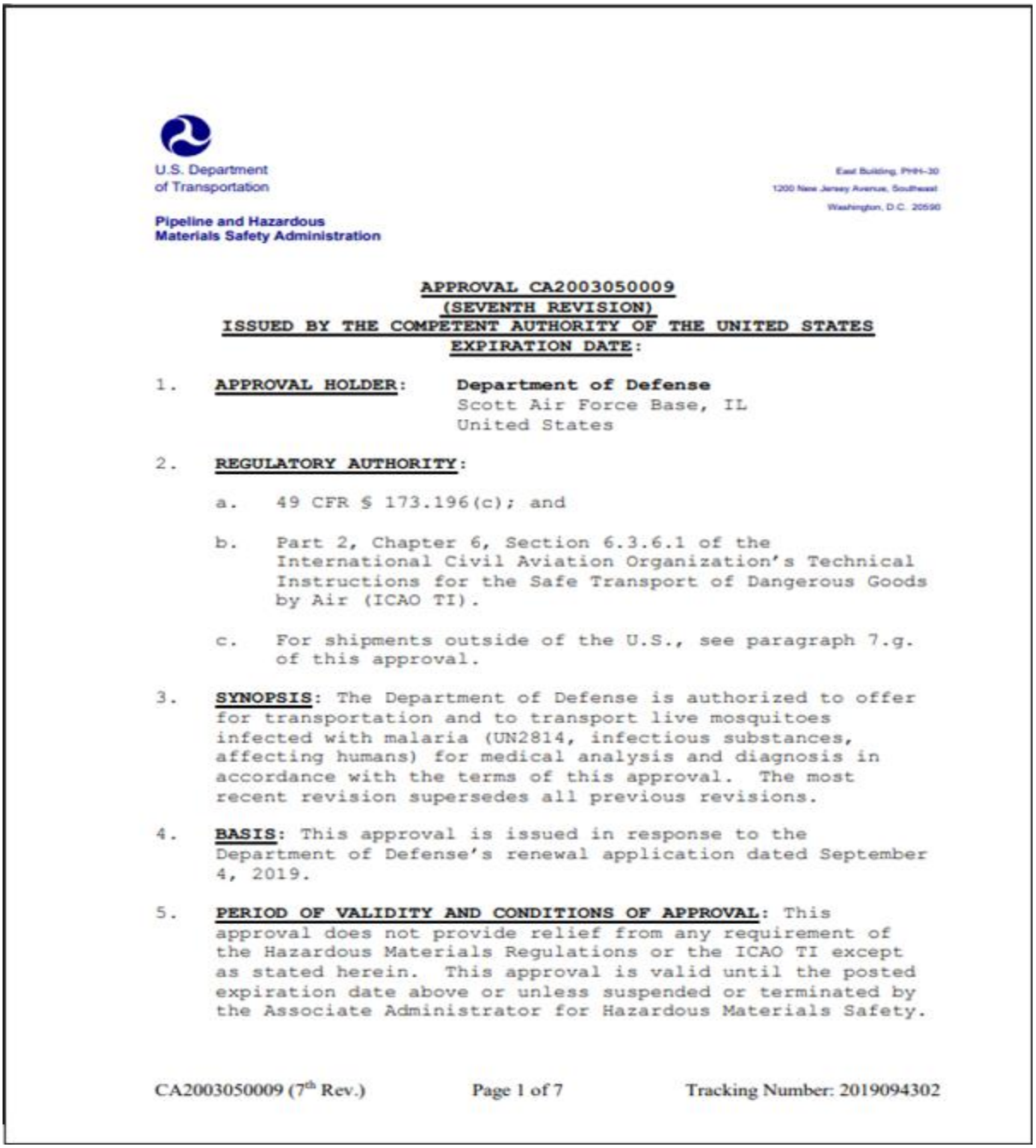
A9.3.1.16. Previous CAA revisions/changes must be included in the package.

A9.3.1.17. Submit a usage report quarterly to the applicable military service/agency focal point, including all CAA renewal requests.

A9.3.1.18. Military service/agency focal point will submit quarterly usage reports to ARTRANS. **(T-0)**

A9.3.1.19. **Figure A9.1 depicts an approved CAA from the Department of Transportation**

Figure A9.1. Competent Authority Approval Sample – Remainder of documents can be viewed on SafetyNet.



U.S. Department of Transportation

Pipeline and Hazardous Materials Safety Administration

East Building, PHH-30
1200 New Jersey Avenue, Southeast
Washington, D.C. 20590

APPROVAL CA2003050009
(SEVENTH REVISION)
ISSUED BY THE COMPETENT AUTHORITY OF THE UNITED STATES
EXPIRATION DATE:

1. **APPROVAL HOLDER:** **Department of Defense**
Scott Air Force Base, IL
United States

2. **REGULATORY AUTHORITY:**

a. 49 CFR § 173.196(c); and

b. Part 2, Chapter 6, Section 6.3.6.1 of the International Civil Aviation Organization's Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO TI).

c. For shipments outside of the U.S., see paragraph 7.g. of this approval.

3. **SYNOPSIS:** The Department of Defense is authorized to offer for transportation and to transport live mosquitoes infected with malaria (UN2814, infectious substances, affecting humans) for medical analysis and diagnosis in accordance with the terms of this approval. The most recent revision supersedes all previous revisions.

4. **BASIS:** This approval is issued in response to the Department of Defense's renewal application dated September 4, 2019.

5. **PERIOD OF VALIDITY AND CONDITIONS OF APPROVAL:** This approval does not provide relief from any requirement of the Hazardous Materials Regulations or the ICAO TI except as stated herein. This approval is valid until the posted expiration date above or unless suspended or terminated by the Associate Administrator for Hazardous Materials Safety.

CA2003050009 (7th Rev.) Page 1 of 7 Tracking Number: 2019094302

Attachment 10

PROCEDURES TO SUBMIT CERTIFICATE OF EQUIVALENCY (COE) REQUESTS

A10.1. General Procedures. Shippers requesting a COE must submit a 'Hazardous Material Data Package' to the Military Services responsible for preparing COEs. A COE is required for domestic and/or international (DoW-owned activity) shipments of a packaged design that equals or exceeds Title 49 CFR requirements. The Air Force COEs are submitted to the Air Force Service focal points for review and concurrence. Other DoW offices will follow their policy procedures with their service focal point and COE POC. The Military Service focal points are identified in **Attachment 7**. Shippers must provide the following information (as applicable), see **Figure A10.1**.

A10.2. Hazardous Materials Data Package. Include the following in the COE package. When the information requested does not apply, write N/A on the line.

A10.3. Example of Completed Package. Include the following in the COE package. When the information requested does not apply, write N/A on the line.

A10.3.1. Requester or petitioner.

A10.3.2. Name.

A10.3.3. Company or activity and location.

A10.3.4. Business telephone number.

A10.3.5. Proposed dates of initial shipment.

A10.3.6. Title 49 CFR provisions.

A10.3.7. Identify all regulatory provisions involved.

A10.3.8. Justify request for COE and specify why DoW and public interest is served by granting a COE.

A10.3.9. Identify why standard provisions of Title 49 CFR are not appropriate.

A10.3.10. Identify how the proposed deviation provides an adequate and reasonable degree of safety.

A10.3.11. Item description.

A10.3.12. Product nomenclature, national stock number(s), part number(s), and EX-number or IHC (if assigned).

A10.3.13. Proper shipping name and UN ID number.

A10.3.14. Chemical Name.

A10.3.15. Common name.

A10.3.16. Hazard classification.

A10.3.17. Form (radioactive materials only).

A10.3.18. Quantity.

A10.3.19. Properties and characteristics.

A10.3.20. Composition and percentage (by volume and weight) of each chemical, if a solution or mixture.

A10.3.21. Igniter ground procedures.

A10.3.22. Net explosive weight.

A10.3.23. Whether or not rocket motor is in a propulsive state.

A10.4. Packaging Data. How the item is packed (drawing showing item/packaging interface) showing any containers, associated fill and relief valves, suspension system, cushioning media, shock indicators, amount of explosives and internal safety features, cutters, dimensions, materials, etc. Drawings must contain enough information to permit engineering comparison between proposed item and the specification requirements or to permit evaluation of the proposed container or shipping configuration. **(T-0)**

A10.4.1. Number of items per inner package/quantity per unit pack.

A10.4.2. Number of inner packages per exterior pack or container.

A10.4.3. DOT specification number for containers (Class 2 or 7).

A10.4.4. Dimensional size of pack and/or container.

A10.4.5. Marking and labeling.

A10.4.6. Container data reflect relevant shipping or accident experience.

A10.4.7. Center of gravity.

A10.4.8. Packaging procedures.

A10.4.9. Test results. State the regulation specifying the tests required and procedures to conduct these tests.

A10.4.10. Previous analogous permits, certificate or approvals.

A10.4.11. Calculations or, preferably, test results of bursting strength and shatter characteristics of pressure vessels.

A10.4.12. Provisions for electrical grounding. **(T-0)**

A10.5. Transportation Description. If the item, as packaged, is a transportability problem item, data required by a transportability report will form a part of this report. Where data is not generated by a transportability report, the following minimal action is required:

A10.5.1. Identify modes of transportation.

A10.5.2. Provide drawings, sketches, or schematics showing different configurations.

A10.5.3. Blocking and bracing.

A10.5.4. Tie-down or restraint.

A10.5.5. Location of center of gravity.

A10.5.6. Consolidation on pallets or/in exterior shipping containers.

A10.5.7. Identify the most probable hazards involved with each handling operation, each mode, or each different type of carrier equipment. Show the need for:

A10.5.7.1. Briefing crews.

A10.5.7.2. Escorts (technical, security police, etc.).

A10.5.7.3. Personnel protective equipment.

A10.5.7.4. Protective environmental equipment/personnel.

A10.5.7.5. Alerting state, military, or government offices of incident or accident.

A10.5.7.6. Exclusive use of carrier equipment.

A10.5.7.7. Specialized materials, equipment, or procedures (non-sparking materials, explosive-proof motors, etc.).

A10.5.7.8. Arms, ammunition and explosives (AA&E) transportation security category if applicable as specified in DTR 4500.9-R.

A10.5.7.9. State what specific action is planned to satisfy each requirement.

A10.5.7.10. Provide reports of tests conducted to verify movement and handling safety.

A10.5.7.11. State what deviations or modifications or Title 49 CFR requirements are needed.

A10.5.7.12. Examples of data item descriptions that may be used to acquire information for substantiating data required in the hazardous materials data package are:

A10.5.7.12.1. DI-L-3311 Explosive hazard classification data.

A10.5.7.12.2. DI-PACK-80880 Transportability report.

A10.5.7.12.3. DI-L-1903 Part, component or subsystem test plan.

A10.5.7.12.4. DI-PACK-81059 Hazardous material POP test report format.

A10.6. Items With No Equivalent Title 49 CFR Packaging. If the item/material has no Title 49 CFR packaging requirements identified on which to base equivalency whether due to size, design, or items with no commercial use, the packaging standard may be based on the following minimum means of evaluation. Paragraphs **A10.6.1 and A10.6.2** include any hazardous material.

A10.6.1. Large packagings not authorized in Title 49 CFR for the item/material to be transported. COEs for large packagings tested in accordance with Title 49 CFR requirements are considered acceptable for most materials if determined to be adequate by service packaging personnel.

A10.6.2. Larger-than-large packagings (LTLP), (packagings that exceed the Title 49 CFR large packaging volume limit of 3 cubic meters). Items/materials requiring packagings exceeding the definitions of Title 49 CFR, and or their components, may have been built to military standards. The packagings themselves were likely designed and built to a military standard and tested to verify its ability to protect and contain the item during shipping and storage. These standards may include:

A10.6.2.1. MIL-STD-331, Fuses, ignition safety devices and other related components, environmental and performance tests;

A10.6.2.2. MIL-STD-648, Specialized Shipping Containers;

A10.6.2.3. MIL-STD-810, Environmental Engineering Considerations and Laboratory Tests;

A10.6.2.4. MIL-STD-1522, General Requirements for Safe Design and Operation of Pressurized Missile and Space Systems;

A10.6.2.5. MIL-STD-1660, Ammunition Unit Loads;

A10.6.2.6. MIL-STD-1791, Designing for Internal Aerial Delivery in Fixed Wing Aircraft;

A10.6.2.7. MIL-STD-2105, Hazard Assessment Tests For Non-Nuclear Munitions;

A10.6.2.8. Industry standards may be acceptable after test methods and results are fully analyzed by service packaging personnel;

A10.6.3. Class 1 items requiring LTLP should be classified as "large and robust" if possible, IAW Title 49 CFR § 173.60(b)(14). Classification as "large and robust" is an explosive classifier function and occurs when the UN Manual of Test and Criteria, Test Series 4 testing is completed on an item (or relevant components), with negative test results. Identification as large and robust allows the most OCONUS shipping possibilities. Container testing required for large and robust items are limited to large packaging as possible, qualification, engineering analysis, and any other tests necessary to protect the asset.

A10.6.3.1. When UN Test Series 4 testing of a Class 1 item is not complete or has not been performed, the classifier may state in an IHC that the item (or relevant components), as part of its operational safety and suitability tests, has been subjected to testing that met the intentions of UN Test Series 4. When the classifier determines the item may be classifiable by DOT as large and robust, container approval for these items may be indicated by packaging personnel issuing a COE based on a selection of the minimum acceptable tests (or proven equivalents) listed in **Table A10.1** below. Additional tests may be required at the discretion of packaging personnel.

A10.6.3.2. If UN Manual of Test and Criteria, Test Series 4 testing was completed with negative test results, then no additional container testing to meet Title 49 CFR §178 requirements is required. The classifier applies for DOT Associate Administrator approval of an FHC, classifying the item as large and robust and describing its container; this is a CAA and its number is used in lieu of a COE number or UN marking on the container.

A10.6.3.3. Larger-than-large packagings (LTLPs), for items that cannot be classified as large and robust, must still have undergone container testing in order to obtain a COE. Packaging personnel will base the COE on a selection of the minimal container tests in **Table A10.1** below. **(T-0)** Additional or alternate tests may be required if indicated by the intended shipping method, environment, or other factors. Required tests to support a COE should be selected by packaging focal points based on the planned transportation modes, configurations, and environments. It is not usually necessary to perform both Title 49 CFR and MIL-STD-648 tests in order to prove a packaging's ability to perform its function, but it may be necessary to perform some combination of both for particular packagings.

Table A10.1. Acceptable Tests-Larger-than-Large Packaging.

49 CFR § 178.950-985 Large package testing	MIL-STD-648 (Qualification and item protection tests)	Comparison of MIL-STD-648 tests to Title 49 CFR tests
Drop test, 178.965, 1 drop on most vulnerable part of container bottom. PG I 1.8m (5.9 ft) PG II 1.2m (3.9 ft) PG III 0.8m (2.6 ft)	- Corner & Edgewise Drop tests, 5.2.4 & 5.2.5, 8 drops at Level A drop height (12 inches). - Transfer-at-sea Shock, 5.2.7, 18- inch flat drop Shipboard shock test, 5.2.9, 18-inch flat drop (when specified in the CIDS). - Conditioned to -20°F or lower, pre-test, if the container, key structural components, or cushioning, is plastic.	- MIL-STD-648 drops not equivalent but is still severe and places more torque and bending stresses on the container base. - Flat 648 drops may be more realistic. - A PG II flat drop on container side or bottom edge may still be required if containers are to be transported in a side-by-side configuration. - Cold conditioning increases test severity for plastic components.
Bottom Lift, 178.970. Loaded at 1.25 x intended weight, ¾ fork penetration: 5-minute hold time. Lift twice, all openings	- Forklift truck (fully captive fork tine enclosures) compatibility test, Para 5.8, Appx P. Loaded at the intended weight. - Forklift truck, (non-captive lift) compatibility test, Para 5.8.1, Appx P. Loaded at intended weight.	- Title 49 CFR lift 2x, all openings; easily performed during any testing. - 648 forklift truck tests more demanding and all (Appx P.5.2, P.5.3, P.5.4 and P.5.4.1) may be advisable unless impractical due to container design or CONOPs.
Top Lift, §178.970 (using base tiedown rings or other designated whole container lifting points). Loaded 2 x capacity Lift and hold for 5 minutes	- Hoisting Fittings, Para. 5.7.3, Appx P. Loaded 5 x capacity lifted 5 min. by intended lifting devices and fittings. - Hoisting Fittings, Para. 5.7.4, Appx P. Planned gross weight, and lift by each individual hoisting point for 5 minutes.	**Note:** - For many/most military containers the tiedowns are also the only available hoisting fittings; containers should not be lifted by devices on the lids unless they are specifically designed and labeled for this.

Title 49 CFR §178.950-985 Large package testing	**MIL-STD-648 (Qualification and item protection tests)**	**Comparison of MIL-STD-648 tests to Title 49 CFR tests**
Stacking §178.980 1.8 x weight of the number of packagings that may be stacked during transportation only. Stacked for 5 min - Test rigid plastic containers at 104°F for 20 days [no RH requirement]	Load test (like containers) [stack test], 5.6.2; S x wt x # of containers that may be stacked to 16 feet (storage), held for one hour; OR 200 x top area x S. (use the largest calculated load). The number of stacked containers may be less if a 16-foot stack is unrealistic in storage. =2 for metal containers; S=1.5 for plastic or other nonmetallic support structure affected by T/RH (and test performed at 120°F/90% RH for 168 hrs (1 wk))	- The MIL-STD-648 stack load is typically much higher, and test period of 1 hour is longer. Load may be lower than Title 49 CFR test if storage stack is only 2 containers high. - The temperature requirement for plastic containers is much higher than Title 49 CFR, & additional RH magnifies the effects of time. - Plastic containers typically fail much sooner than 20 days in these conditions (usually 1-24 hours).
Vibration §178.985 Vertical or rotary double amplitude peak to peak displacement of 1” for one hour	Repetitive Shock, Para. 5.2.2 (ASTM D999, Method A1 or A2)	Both methods are the same
	- Resonance survey & dwell, 5.3.3.1 - Random vibration, Para. 5.3.2 - Sinusoidal cycling test, Para, 5.3.4	- Resonance dwell is the most severe vibration test of the shock system (cradle, coil-mounts, cushioning, possibly the base, etc.) - Random vibration may be more severe depending on selected profile.
None	Pendulum Impact tests, 5.2.6: - Truck transport, 5.2.6.1, using 5 ft/s or 7 ft/s - Rail transport (containers over 9.5 ft long, using 11.88 ft/s - Conditioned to -20°F or lower if plastic	- Not directly comparable to any Title 49 CFR test but more realistic for truck and rail ground transport. Both item and container must pass. Depending on container size, impacted on shortest side. Typically, only the base wall is impacted, sometimes the lower portion of the lid wall also; the smaller impact area concentrates the force of the impact. Tests restraint of the item. **(T-0)**

A10.7. COEs prepared by Service packaging personnel shall provide the following information. Hard copy COEs must accompany shipments of its subject item/container, along with referenced IHCs, FHCs, DOT-SPs, or CAAs. An example is noted in **Figure A10.1** below. **(T-0)**

A10.7.1. CCN.

A10.7.2. Authority.

A10.7.3. Issued by, with signature.

A10.7.4. Basis for certification (test reports and or engineering analyses (by reference)).

A10.7.5. Packaging description.

A10.7.6. Mode(s) of authorized transportation.

A10.7.7. Interim Hazard Classification, when applicable.

A10.7.8. DOT-SP by reference when applicable; **Note:** - if the DOT-SP authorizes the packaging, the COE is not necessary.

A10.7.9. CAA(s), when applicable.

A10.7.10. DOT hazard classification (Proper Shipping Name, Label, Marking).

A10.7.11. DoW hazard classification (Security Classification Guide, Division, UN Identification Number) (if different than DOT information).

A10.7.12. Expiration date The Military Service/Agency that authorizes COEs may specify expiration date on COEs.

A10.7.13. Requirements of this Attachment are considered minimum essential information to substantiate issuance of a COE. When any of the information changes or is revised, the COE must be amended and re-submitted for approval at the time such information becomes available. Any changes to a COE requires that a new COE be approved and issued. **(T-0)**

Figure A10.1. Certificate of Equivalency Sample - Remainder of documents can be viewed on SafetyNet.

DEPARTMENT OF THE AIR FORCE
406th SUPPLY CHAIN MANAGEMENT SQUADRON (AFMC)
HILL AIR FORCE BASE, UTAH

OO-ALC CONTAINER CERTIFICATION

AF 13-251
REVISION SIX

1. **AUTHORITY:** Certification of Equivalency (COE) is issued pursuant to Title 49 CFR 173.7(a) of the Department of Transportation (DOT) hazardous materials regulations and authority established in DAFMAN 24-210, Packaging of Hazardous Material.

2. **COMMODITY:** Rocket Motor, Medium Range Ballistic Missile (MRBM) Booster. Item configuration consists of one Minuteman II Stage 2 SR19 rocket motor (ANB-3066 propellant), one Minuteman III Stage 3 SR73 rocket motor (ANB-3066 propellant), and an ordnance destruct system.

NSN or P/N	EX Number or IHC	N.E.W (Kg)
P/N 1096-2001	H24-145 (or Current)	20931.300 lbs (9494.278 kg)

DOD Hazard Class/Division/SCG/SG: 1.3C
DOT Proper Shipping Name: ROCKET MOTORS
UN Serial Number: UN0186
DOT Label: EXPLOSIVE 1.3C

3. **BASIS:** 49 CFR 172.102 SP 109, Rocket Motor must be non-propulsive in transportation. 49 CFR 173.62 UN0186 PI 130, Larger than Large, 177.835 (i)(j)(k) motor vehicle transport, Items requiring Special Approval. This asset has shipped since 2013 without incident. Packaging method is proven safe and effective.

4. **PRESCRIBED PACKAGING:** Asset will be placed in its appropriate carriage(s) by part number in accordance with procedures in T.O. 21M-LGM-30B-2-26. Tie down procedures will be in accordance with T.O. 21M-LGM-30B-2-26 or T.O. 35M4-3-6-1. The carriage(s) with Motor(s) will be transported in a multistage transporter NSN 1450-01-139-0286AH or NSN 1450-01-468-1093AH or the rocket motor semitrailer P/N 8H1E-10, NSN 1450-01-345-9693AH using standard procedures in effect for operational motors and carriages. Refrigeration, temperature control, will be utilized at all times while the asset is in the trailer. This Packaging meets or exceeds the requirements of 49 CFR.

5. **MODES OF TRANSPORTATION:** Motor Vehicle (Reference specific HAZMAT Regulations governing mode being used for any updates, additional limitations, or restrictions).

6. **SPECIAL PROVISIONS:**

   a. This COE is restricted to the commodities listed above.

Attachment 11

PROCEDURES FOR ADDING HAZARDOUS MATERIALS TRAINING CLASSES

A11.1. General Procedures. All personnel involved with the preparation and shipment of HAZMAT for transportation must receive training in accordance with DTR 4500.9-R, Part II, Chapter 204; 49 CFR §172.704, Training Requirements; and DoW Component/Service/Agency HQ regulations. Training for military air shipments must be in accordance with AFMAN 24-604. In addition, Title 49 CFR §172.704 states that there are pre-transportation function specific, safety and security training requirements that DoW personnel must meet. **(T-0)**

A11.2. Approved Schools. Schools, other than those identified in the DTR, are not authorized to provide instruction on behalf of an approved training school. Delegation of training development and delivery of instruction requires submission of formal written request through the Service Focal Point and approval by the HMPWG. Transfer of Program of Instruction (POI), for the purpose of delivering instruction to a school not approved in the DTR is not permitted. Schools and courses published prior to the date of this publication are excepted based on prior criteria, including technical specialist courses.

A11.3. Evaluation Process. This process defines the requirements for DoW entities to submit requests to provide an additional resource for the current 80-hour HAZMAT training course. It also outlines the process for new technical specialist or function-specific training to be used in place of the current 80-hour HAZMAT training class as allowed in the DTR training. Each request should provide justification regarding why the course is required, the business case to support the justification and the administrative process that will be used to track and verify registration, and training, testing and record retention. If an additive training provider of the 80-hour HAZMAT course is approved, it will be added to the DTR and available for all DoW personnel to attend. The school/source must be willing to be a signatory to the existing DoW Memorandum of Understanding (MOU) managed by the Hazardous Materials Training Working Group (HMTWG). If the new course is an alternate to the current 80-hour on-site certification course, the Service Focal Point must maintain oversite of the course and review/verify its content every five years. The DoW Hazardous Material Packaging Working Group (HMPWG) is the official reviewer and approving entity for all proposed courses. **(T-0)**

Table A11.1. Course Justification/Approval.

Command Name/POC		
Title of proposed training course		
Title 49 CFR §172.704 - Define the training requirement General awareness/familiarization training Function-specific training Safety training Security awareness training In-depth security training		
Explain why the new course is required and provide a Business Case Analysis to meet the requirement.		
Explain why current courses do not meet the training need (i.e., limited availability of existing course seats, specific process not detailed in existing course, limited shipping requirements, location/commodity specific requirements, others, etc.)		
Pre-evaluation/approval of the course - Provide copies of the below documents		**Check if included**
Program of Instruction (POI)		
Practical exercises and how they are presented		
Handouts that will be provided to the students		
Course Sustainment - Provide copies of the below documents		
Process to update the POI and the intervals this will be done		
Plan for instructor training/retention going forward		

A11.3.1. Please complete the above and allow four (4) months for the HMPWG members to review. If the content and justification is approved, a pilot class will be conducted and attended by HMPWG members or designated subject matter experts. **(T-0)**

A11.3.2. If a pilot course for the 80-hour HAZMAT course is approved, the requested Command funds the travel costs of HMPWG members/Subject Matter Experts to attend and review the pilot course. A 60-day notice of the class dates should be given to the HMPWG in order to ensure a majority of the members are able to attend. The HMPWG provides any deficiencies to the requesting Command for correction within 30 days following the pilot. The requesting Command has 30 days to make those corrections. The HMPWG recommendation for approval or disapproval is presented to the Defense Packaging Policy Group (DPPG) within 30 days after the corrections are reviewed. If approved as an additional provider of the 80-hour HAZMAT course, the provider is added to the DTR.

A11.3.3. If a pilot course for a new technical specialist or function-specific training being used as an alternative to the current DTR training course is approved, the requested Command funds the travel costs of HMPWG members/Subject Matter Experts to attend and analyze the pilot course. The number of members is determined based on the course and content. The HMPWG provides any deficiencies to the requesting Command for correction within 30 days following the pilot. The requesting Command has 30 days to make those corrections.

A11.3.4. Each school that is authorized to provide HAZMAT training, will provide the HMPWG with the number of students trained (by DoW Component), failure rate, student course critique rating, list of new and current instructors (to include credentials), and a POI updates annually. Every five years the Command will provide funding for a HMPWG members/Subject Matter Experts member to re-evaluate the course. **(T-0)**